

SELF- ASSESSMENT QUESTIONS ON STEELS

- (a) Classify the steels as a function of chemical composition and for each associate a suitable application in terms of their distinct properties.
- (b) Cite three reasons why ferrous alloys are used so extensively in a wide range of applications
- (c) Cite three characteristics of ferrous alloys that limit their utilization.
- (d) Explain the advantages and disadvantages between strengthening by CW vs heat treatment in steels in relation with %C
- (e) Explain how strength and toughness are balanced in CW and heat treated steels to meet application needs
- (f) Cite the 4 main causes of embrittlement in steels and their counter measure
- (g) Explain the difference between MnS and oxides inclusions in steels and their influence on mechanical properties
- (h) Define hardness and hardenability and make an evidence of both in the Jominy test (hint: depict the Jominy curve)
- (i) Explain how and why hardenability affects TTT curves
- (j) Present the main chemical reactions occurring in a blast furnace
- (k) Cite 3 factors explaining why the integral production of steels is made by way of cast (pig) iron, rather than from steels
- (l) Explain why and for which kind of loading conditions (e.g., bending, shear, torsion, tensile) hardenability is crucial
- (m) Explain the influence of tempering on microstructure and mechanical properties
- (n) Explain the strengthening mechanisms and factors influencing the mechanical properties of HSLA and 38NiCrMo4 steels
- (o) Compare the metallurgical phenomena on continuous cooling of hypoeutectoid steels with the aid of CCT on:
 - a. quenching in a water bath and b. moderately severe cooling bath such as warm oil bath

SELF ASSESSMENT QUESTIONS ON PRODUCTION OF STEELS - LECTURE 10b

1. Classify Steels based on the productions routes based on grade quality, mass production, cost, degree of recyrcing
2. Explain why steels are produced from pig iron
3. Explain the main furnace operations in the blast furnace
4. Explain and justify the relevant chemical reactions in the blast furnace
5. Illustrate the most effective reducing agent of iron ores in the blast furnace
6. Explain the composition and roles of limestones
7. Comment on the function of the BOF
8. Justiy the detrimental role of S and how its content is reduced in the melt
9. Illustrate how atmospheric reactions at the melt are minimized during transfer of liquid from a station to another
10. Compare the roles of VOD and AOD and steel destination
11. Explain why H is detrimental in steels and how it can be eliminated
12. Explain what are the most effective deoxidizing elements in a steel melt and their consequence
13. Explain why N has an harmful role and in which steels
14. Describe how degassing is performed in a steel melt and to which impurities it applies
15. Explain the TMCP metallurgical phenomena affecting the mechanical properties on furnace heating of HSLA Nb-steel plate
16. Describe the main purpose of the TMCP in the production of HSLA Nb-steel strips
17. Comment on the TMCP metallurgical phenomena after finish rolling of Nb-HSLA steel strips
18. Correlate the interplay between metallurgical phenomena and mechanical properties at coiling of Nb-HSLA steel strips
19. Explain the consequences on mechanical properties imparted by hard-brittle and soft-plastic inclusions in steels

- i. Specify the main concentration limits in steels to discern alloyed from unalloyed steels (AND THE LIKE)
- ii. Provide the alphanumeric code for a structural steel of min yield strength of 550 MPa, min impact energy of -30 J at -30 °C (AND THE LIKE)
- iii. Specify the alphanumeric code for a tempering unalloyed steel (< 1% Mn) of average %C =0.25 (AND THE LIKE)
- iv. Specify the alphanumeric code for a tempering unalloyed steel (< 5% alloying elements) of average C =0.42%, Cr=1% and present but unspecified Mo e S (AND THE LIKE)
- v. Specify the alphanumeric code for a tempering alloyed steel of average C =0.38%, Cr=5% , Mo=3% and present but unspecified V(AND THE LIKE)
- vi. Specify the alloying concentrations of AISI 4140
- vii. Specify the alloying concentrations of AISI 4340
- viii. Specify the alloying concentrations of AISI 4620
- ix. Specify the alloying concentrations of AISI 1020 (AND THE LIKE)
- x. Specify the AISI designation of a Ni steel.